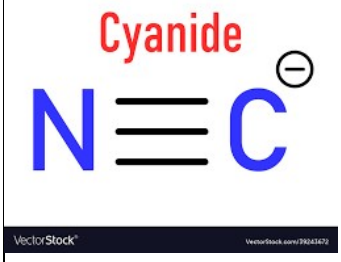
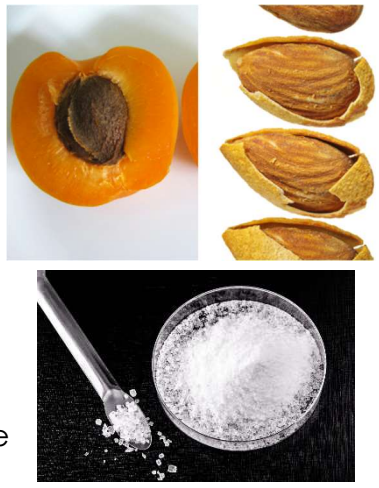
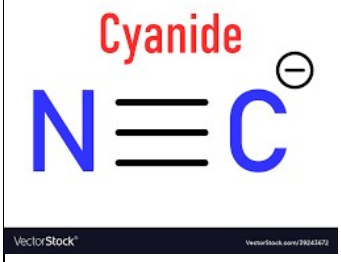
Cyanide





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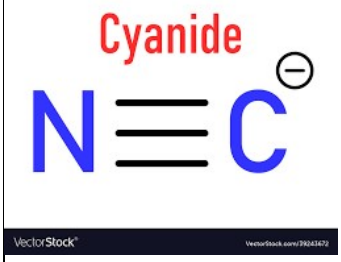
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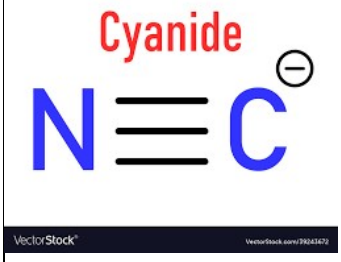
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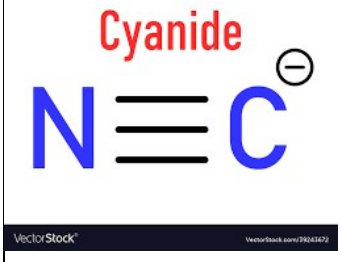






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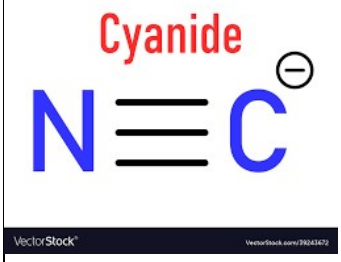






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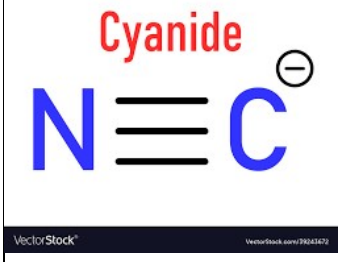






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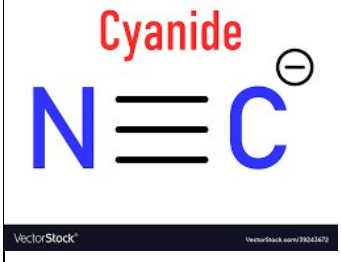






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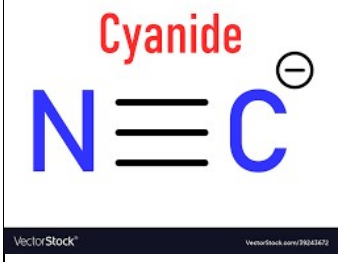






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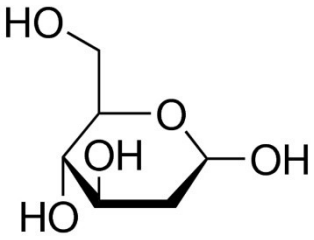




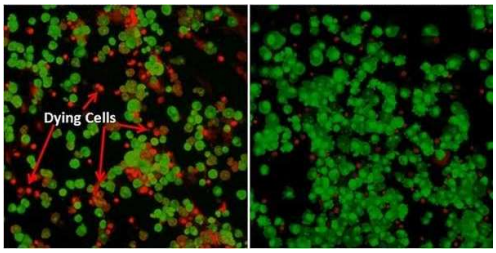


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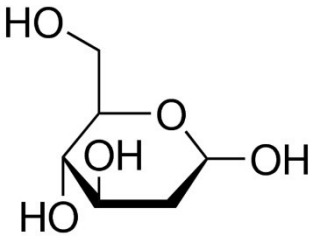
2-deoxyglucose



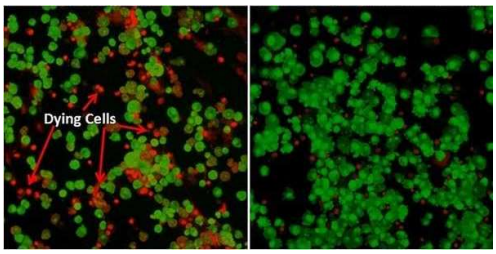
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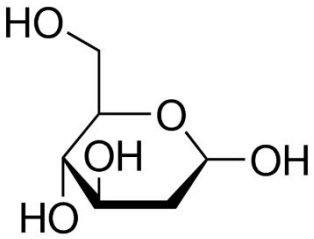
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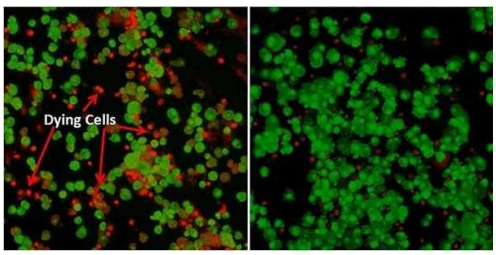
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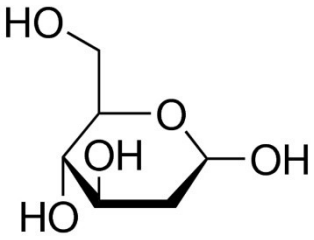
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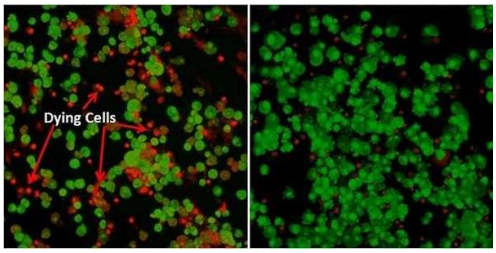
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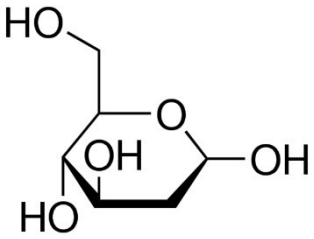
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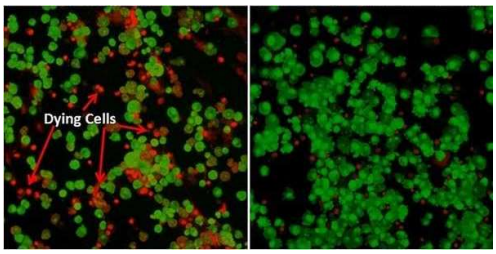
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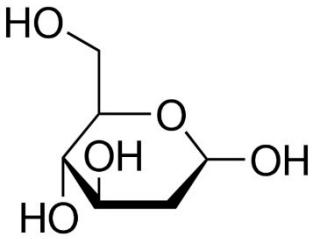
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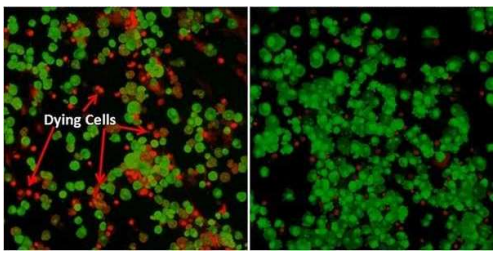
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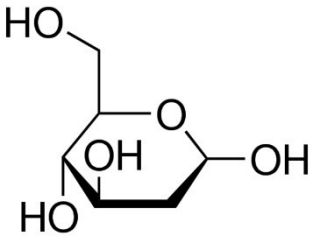
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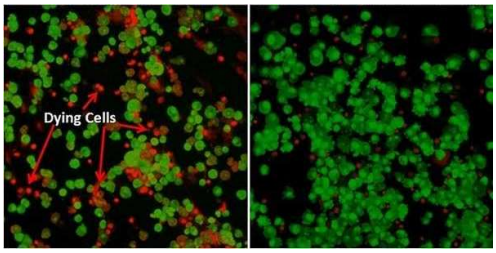
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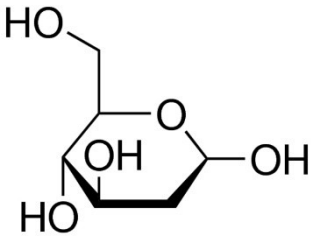
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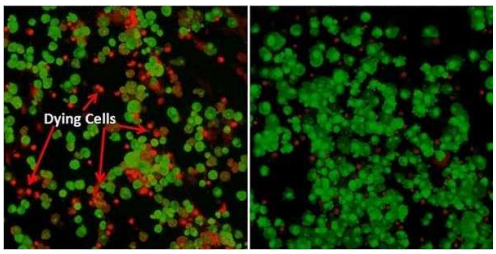
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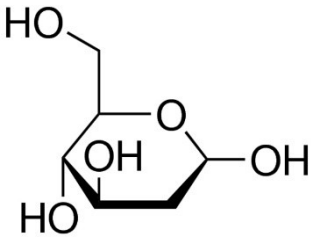
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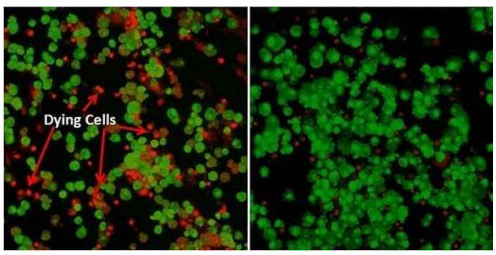
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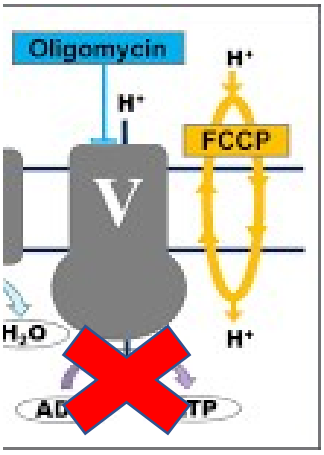
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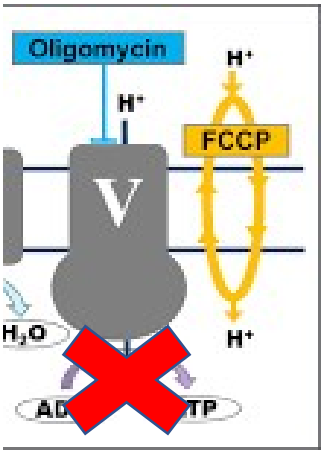
Oligomycin



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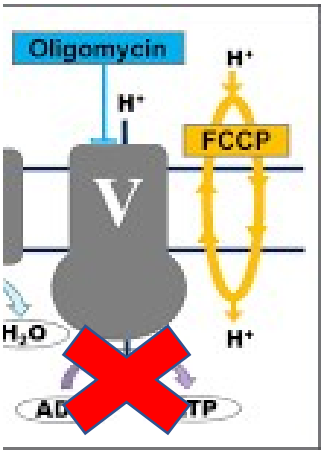
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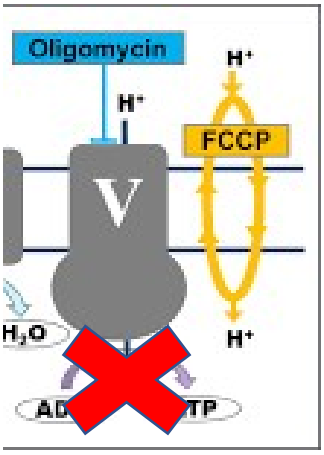
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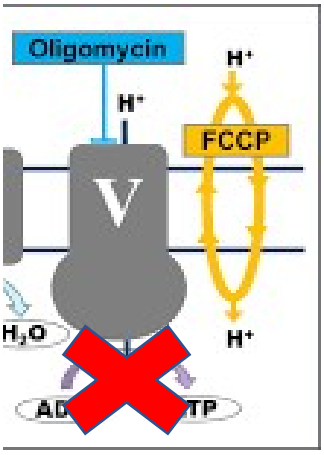
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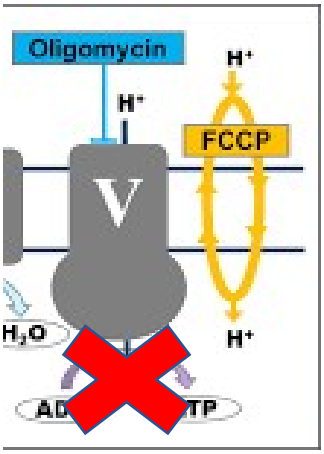
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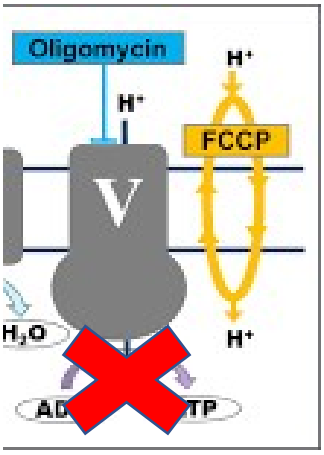
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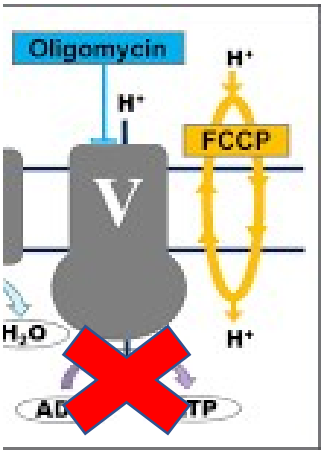
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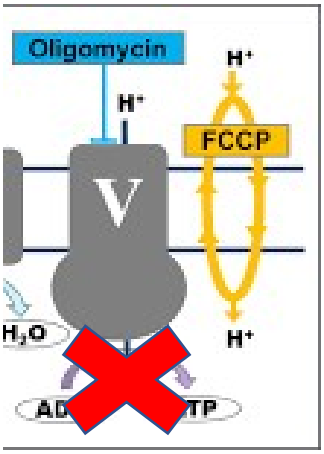
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Oligomycin was first found in 1954 isolated from a bacterium that belongs to the streptococcus family. It was found to induce cellular death by interrupting the Electron Transport Chain of cellular respiration. This protein binds to ATP synthase and prevents protons (H+) from being pulled back down from the intermembrane space. Without the H+ flowing through the ATP synthase, it cannot convert ADP into ATP. For a short time, this compound was used in nerve agent in gas during war and in pesticides. It is no longer used.



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Arsenic



Arsenic is a naturally occurring element on Earth. It is usually found in combination with other minerals but can be by itself in crystal form. Arsenic is naturally found in the Earth's crust. Arsenic is considered to be a very potent poison. When ingested or absorbed through the skin, arsenic enters the mitochondria of cells. It attaches itself to the first enzyme in the Krebs cycle. This prevents pyruvate from entering the Krebs cycle at all, essentially interrupting cellular respiration in the middle of the process. Arsenic is currently used in car batteries, some ammunition, and some pesticides. It used to also be used in a yellow color that was popular in Victorian times and could be found in wallpaper around the world. Currently the FDA is trying to have arsenic banned from pesticides. Interestingly there are Bacteria on earth that can use arsenic in their cellular processes.



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Case Study #1: Tim Braxton

Note: This case study is based on REAL events, however names and circumstances have been changed.



On Halloween in 1974, Eight year-old Tim Braxton was brought to the hospital after ingesting Halloween candy, specifically Pixy Stix. Pixy Stix are a powdery candy that comes in a tube. He was eating candy but noticed that it tasted bitter. After telling his parents that the candy tasted weird, he ended up passing out and foaming at the mouth. When he presented to the hospital, he has muscle weakness and his heart stopped beating twice. Police were called and it was found that his candy had been laced with a poison by a neighbor. He died of heart failure later that evening.

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Case Study #2: Mei & Genji Nakamura

Note: This case study is based on REAL events, however names and circumstances have been changed.



In 1990, a woman living on a farm in Japan was found unconscious on her front porch. She was rushed to the emergency room at a local hospital where she was found to have muscle weakness and heart palpitations. Eventually she lost control over her respiratory muscles, leading her to be intubated. She finally died by cardiac arrest. Her doctor suspected exposure to a toxic nerve agent. When police investigated her home, they found a pesticide she had been using on her vegetables. This pesticide contained a deadly toxin. When ingested, this pesticide can lead to rapid death.

Case Study #2: Mei & Genji Nakamura

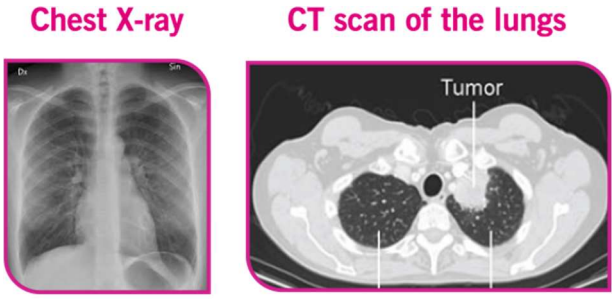
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Case Study #3: Vanessa Craft

Note: This case study is based on REAL events, however names and circumstances have been changed.



In 2003, Vanessa craft was getting treatment for lung cancer when she went into sudden cardiac arrest. Vanessa had been getting an experimental new treatment for her cancer. Her treatment plan included injecting the tumor cells with a poisonous compound. By focusing it on the cancer cells, her doctors were able to start reducing the tumor's size by killing the cancel cells. Unfortunately, a tired doctor mistyped the dose and Vanessa was injected with 10x the normal dose of the poisonous compound. 20 minutes later she went into cardiac arrest and died.

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Chest X-ray CT scan of the lungs Tumor

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Case Study #4: 20 Malaysian Children

Note: This case study is based on REAL events, however names and circumstances have been changed.



In 2005, an elementary school in Malaysia was rocked by tragedy. At least 20 school children were sent to the hospital after eating a snack of cassava, a root that is poisonous if you do not prepare it correctly. Eaten, raw the human digestive system can convert part of the root into a deadly poison. The students stopped eating the root because they said it tasted bitter. 10 minutes later they complained of stomach pain and started vomiting. Many complained about weakness and dizziness. Because the nearest hospital was 20 miles away, the children died of respiratory collapse en route to the hospital.

Case Study #4: 20 Malaysian Children

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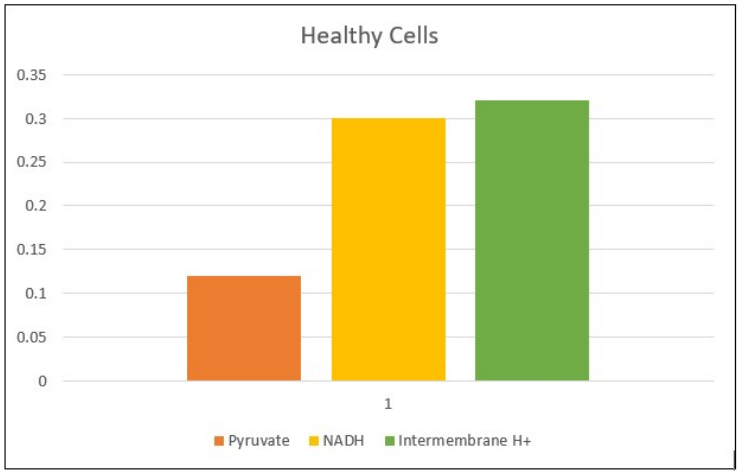
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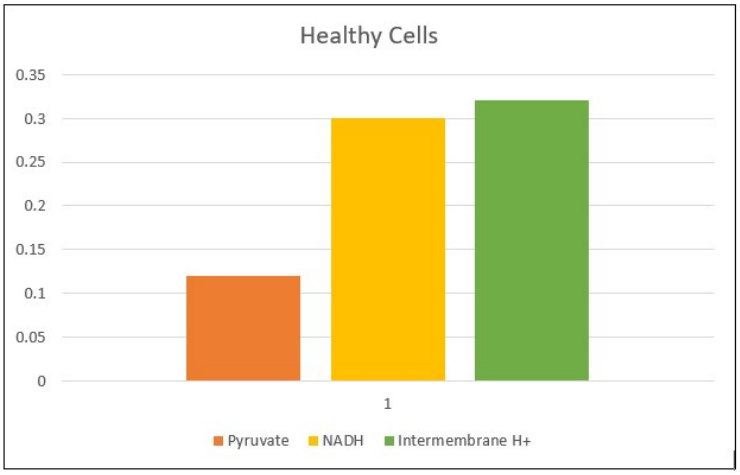
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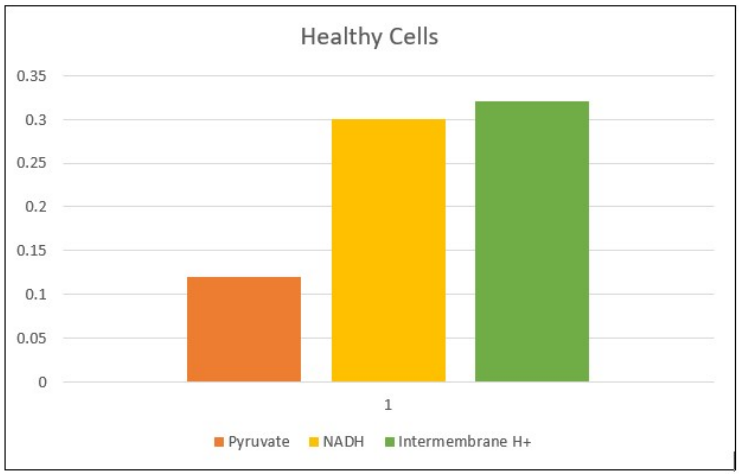
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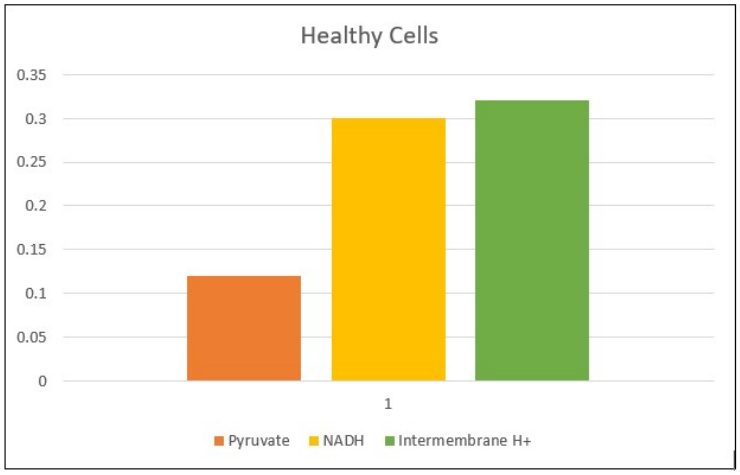


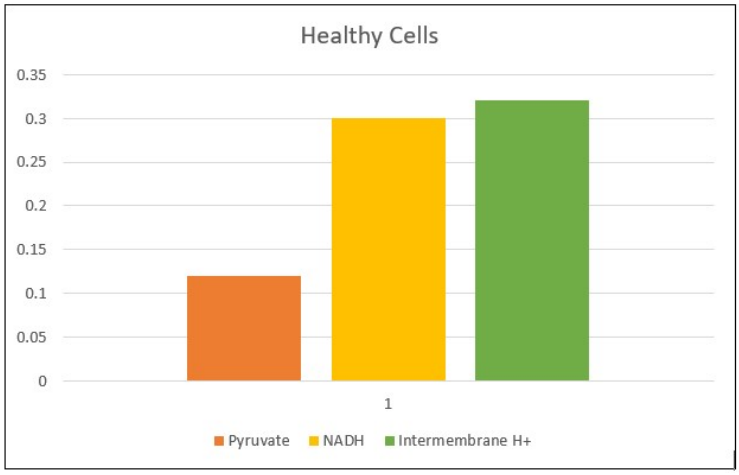
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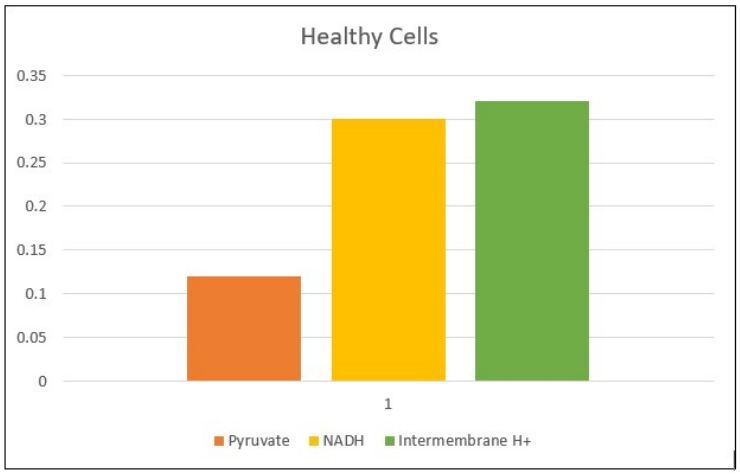


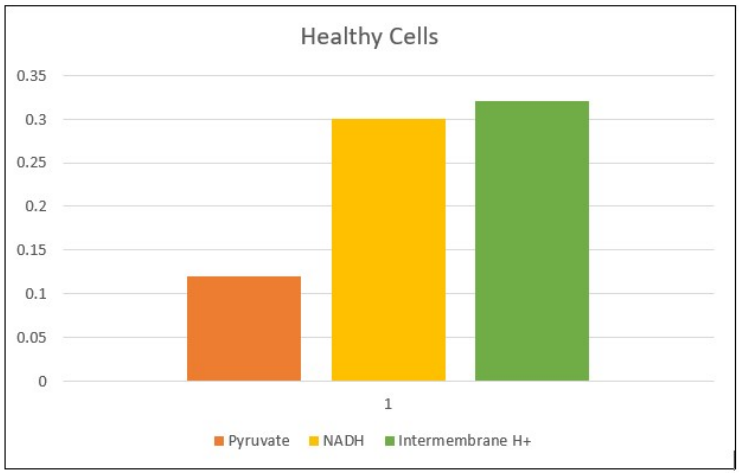


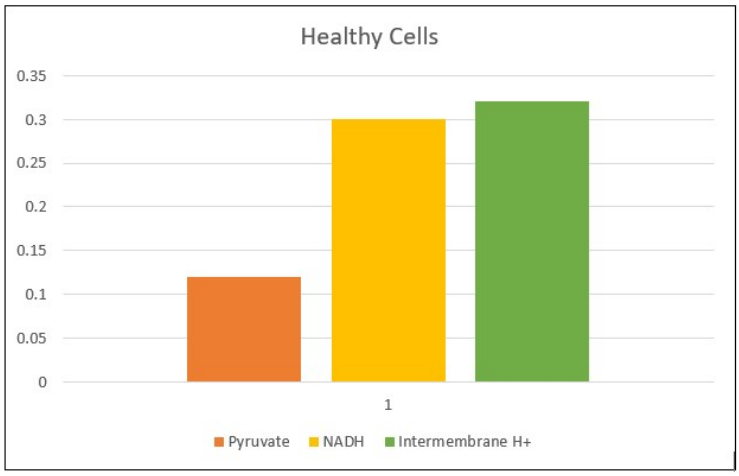


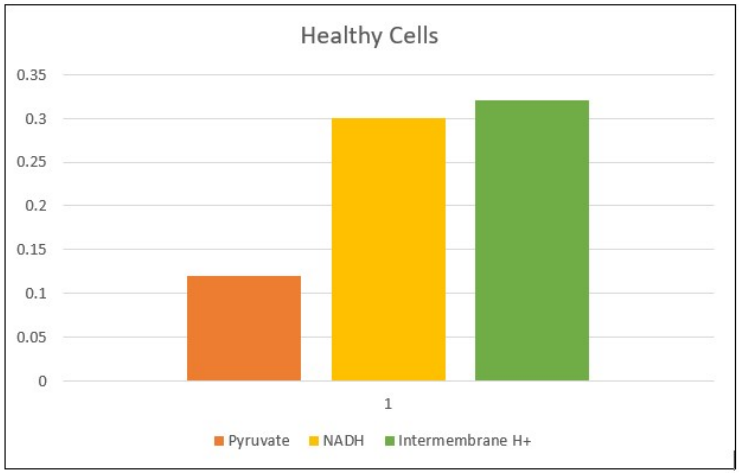


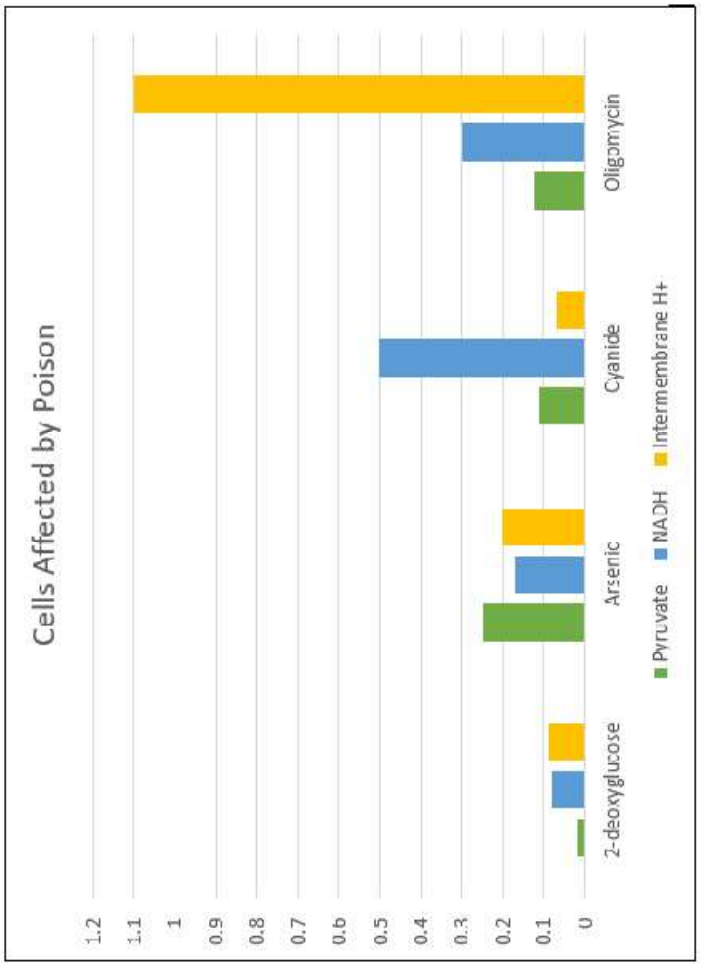


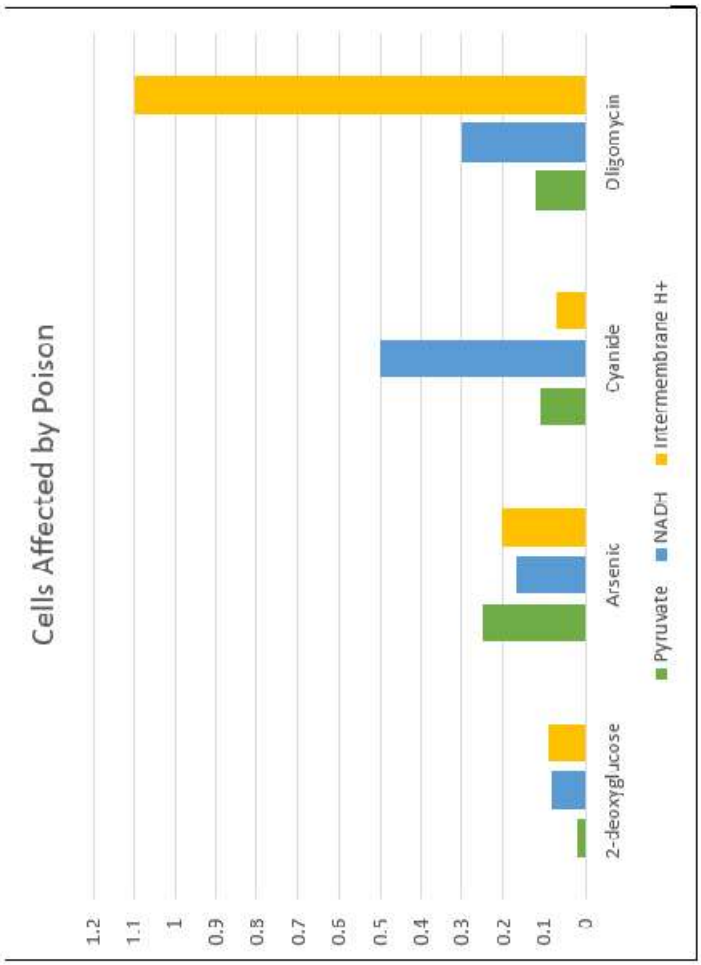


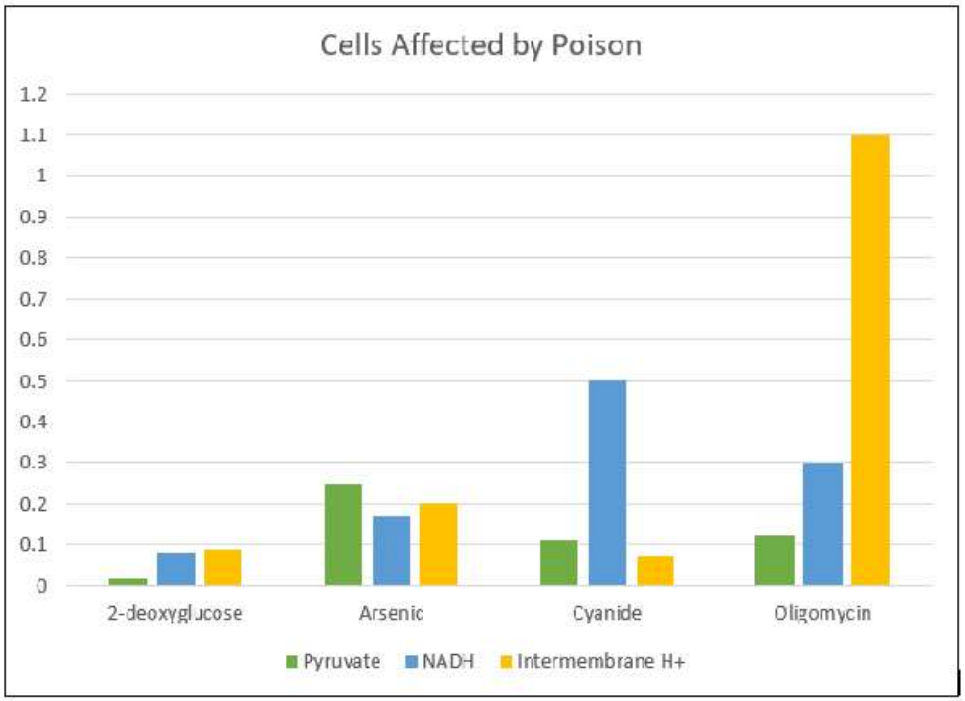

Healthy Cells
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
1
Pyruvate
NADH
Intermembrane H+

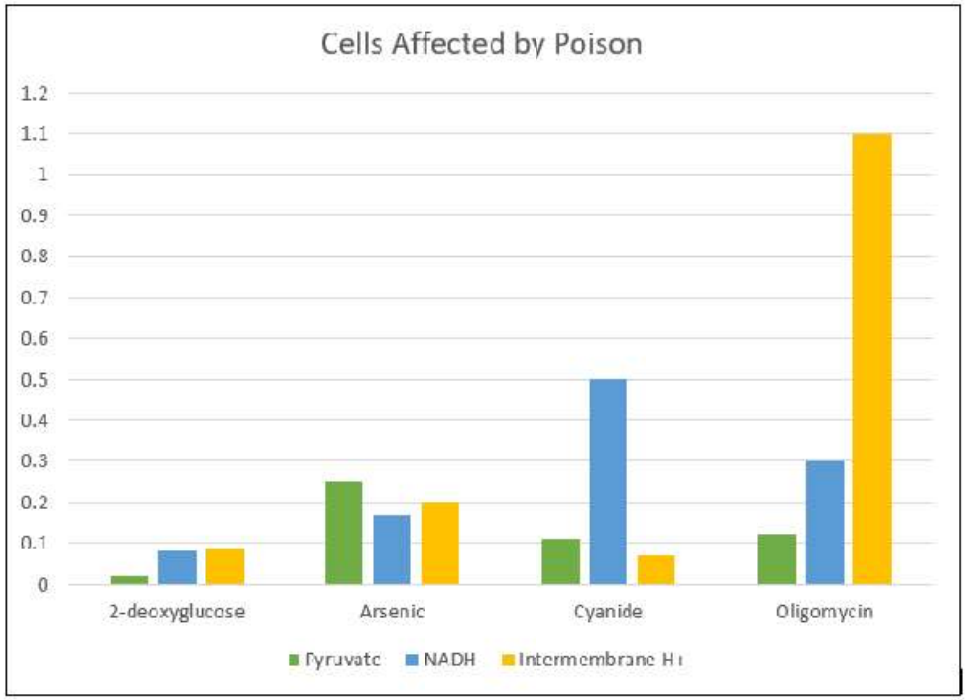

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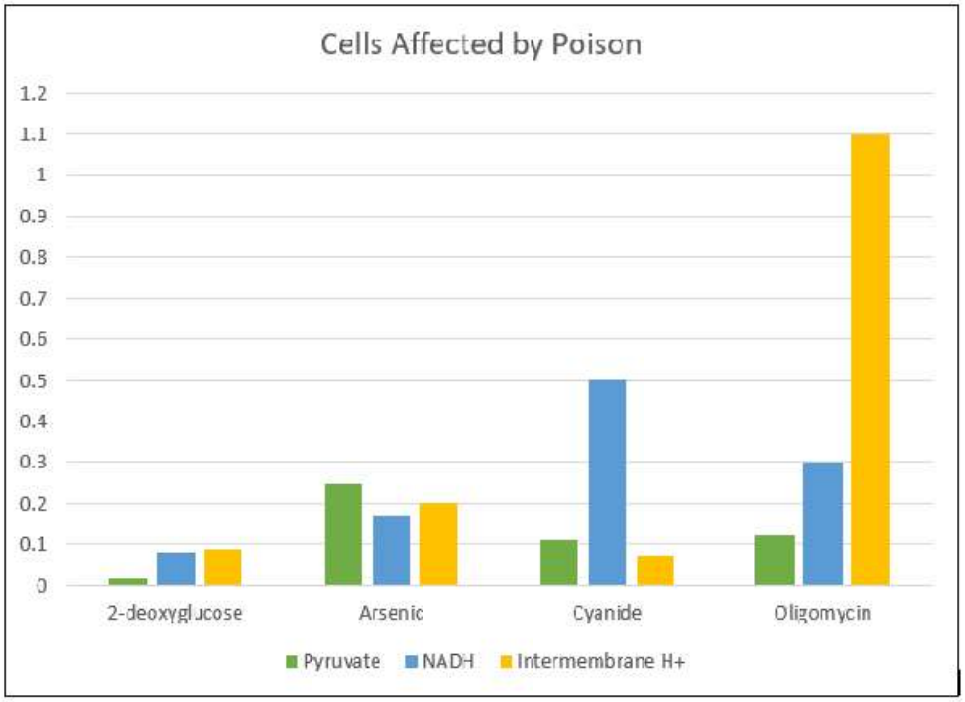

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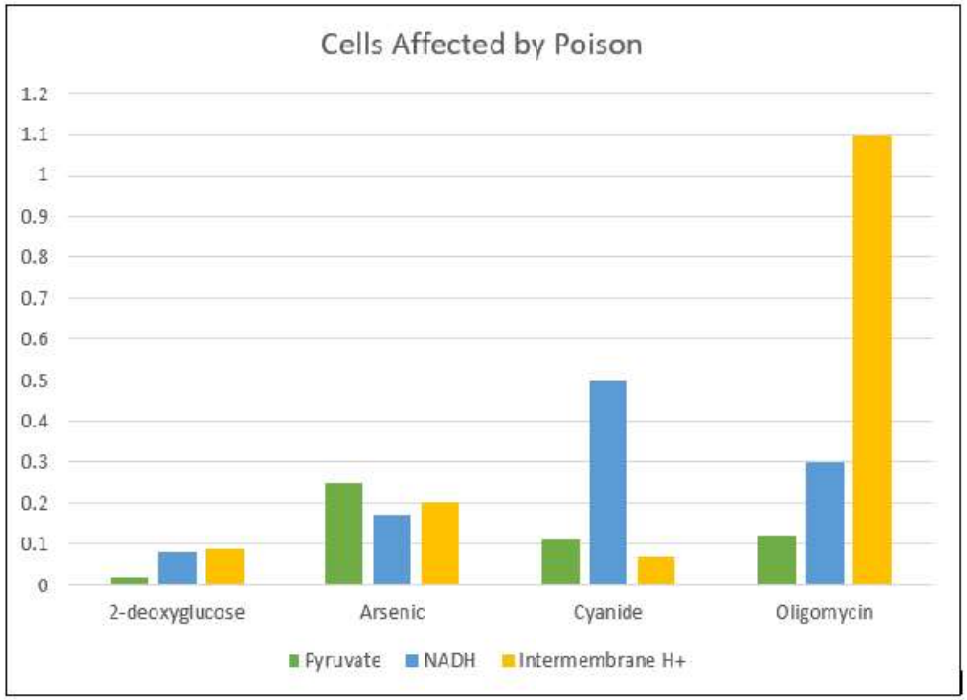

Cells Affected by Poison
Oligomycin
Cyanide
Arsenic
2-deoxyglucose
Pyruvate
NADH
Intermembrane H+
1.2
1.1
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0

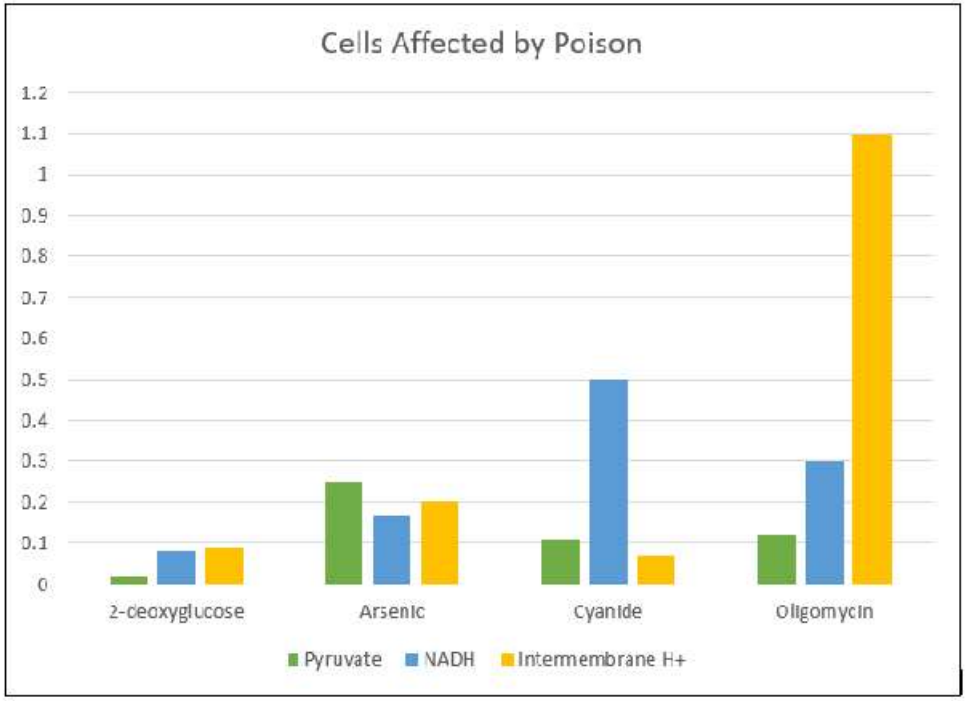

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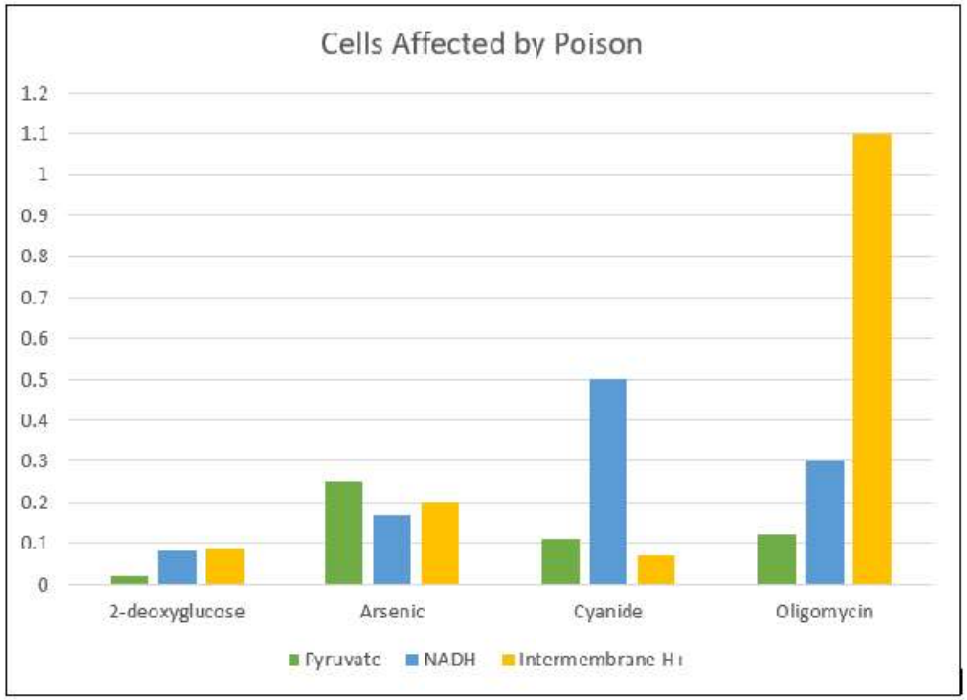

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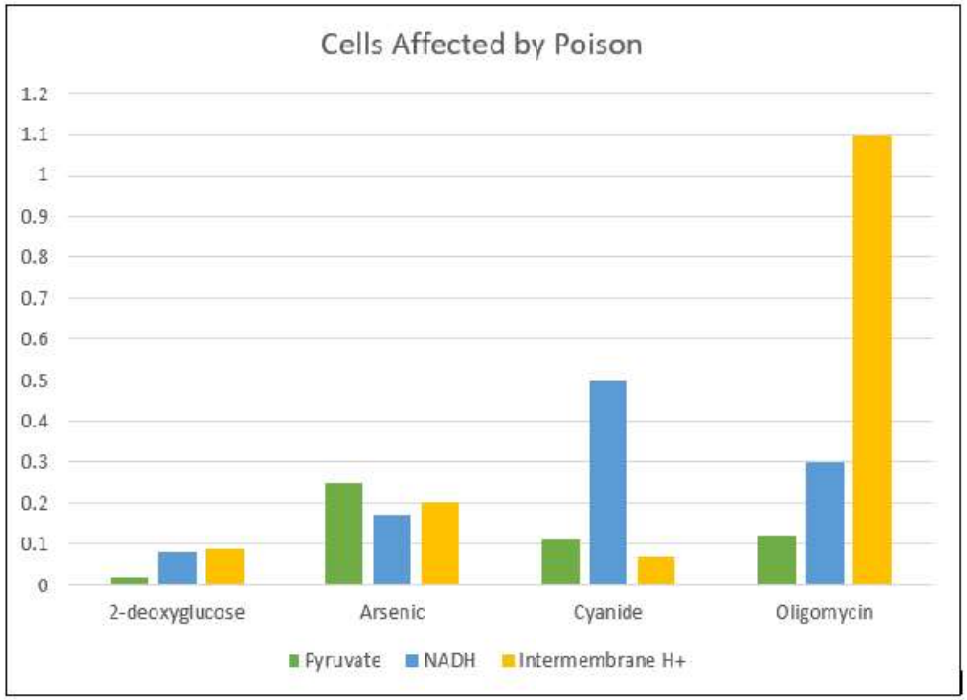

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